Computer Organization and Architecture
(Code: CS256)

Lecture-28: External Memory Design

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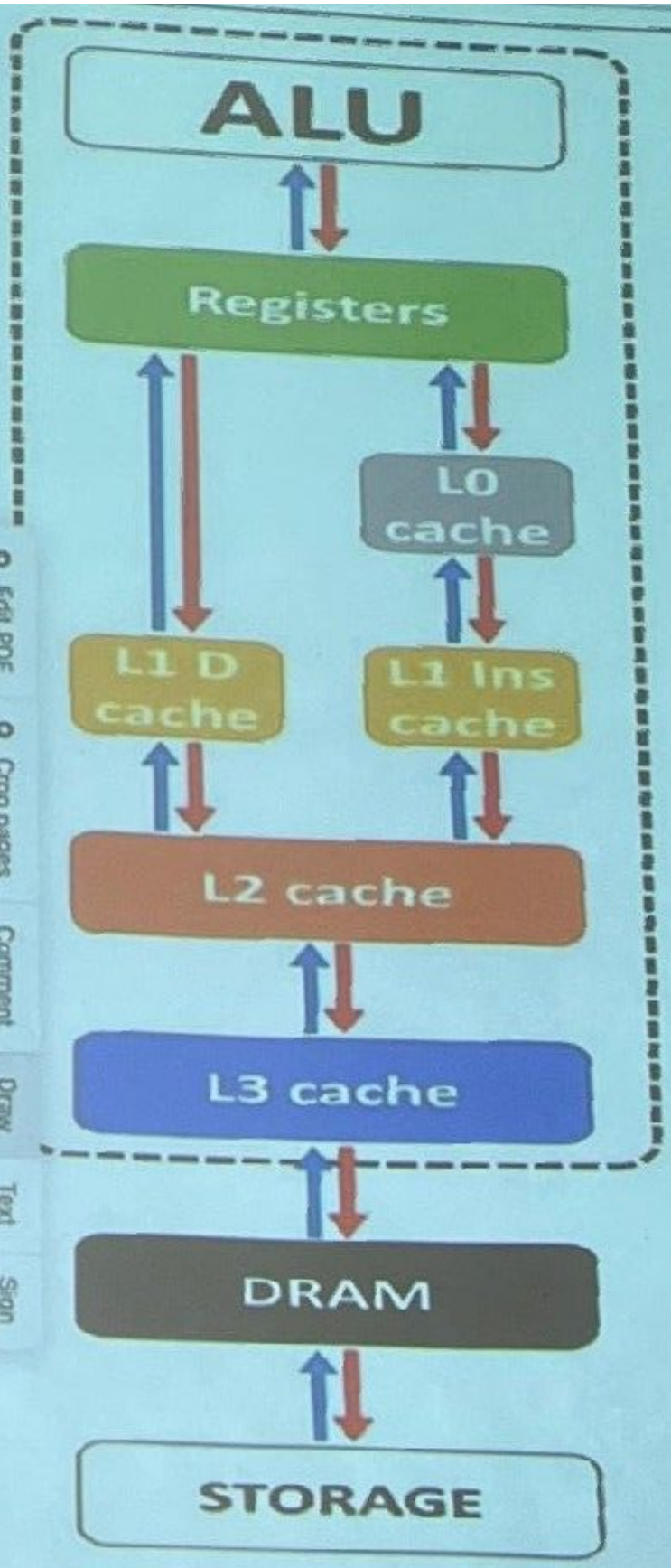

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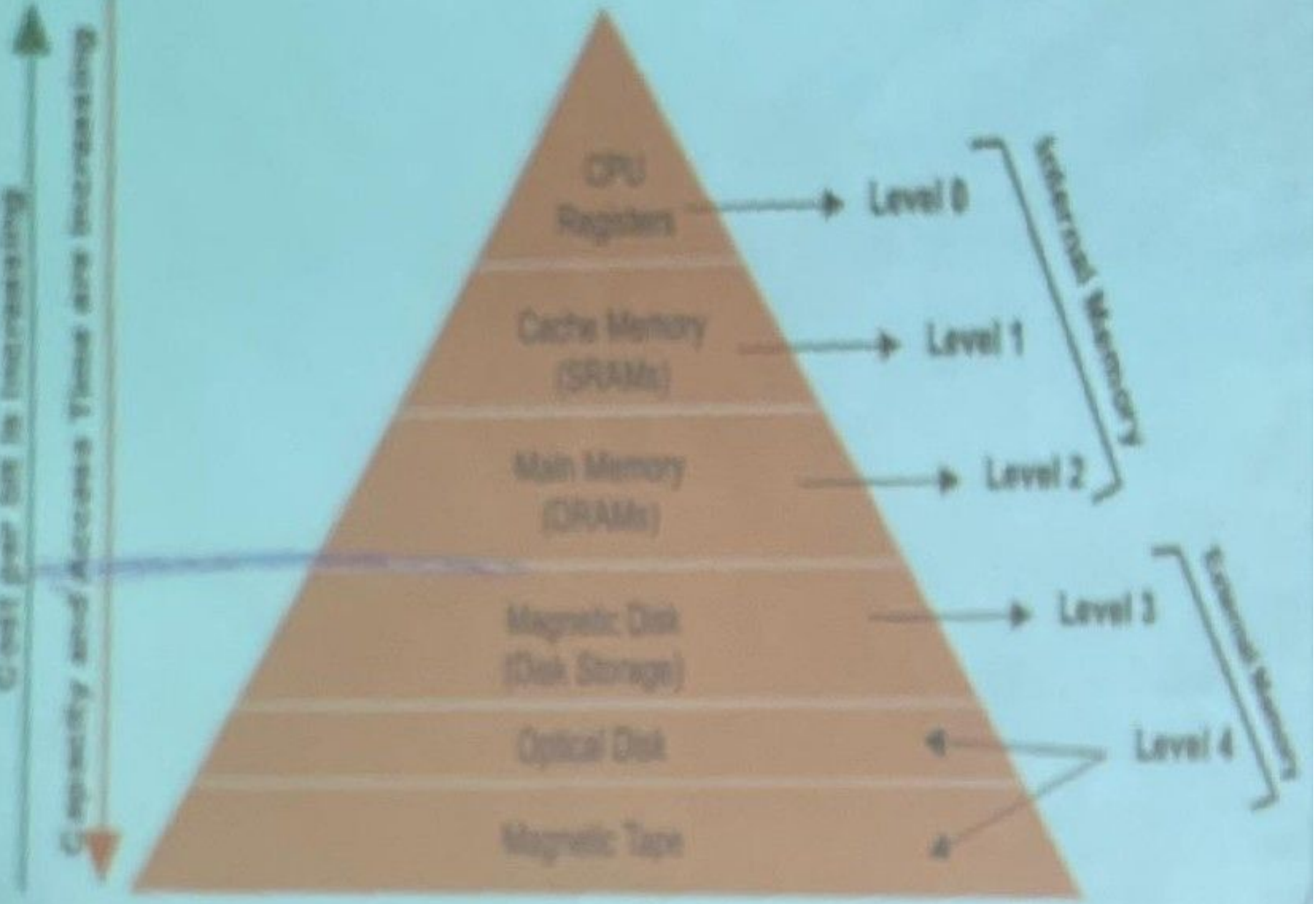
Memory Hierarchy

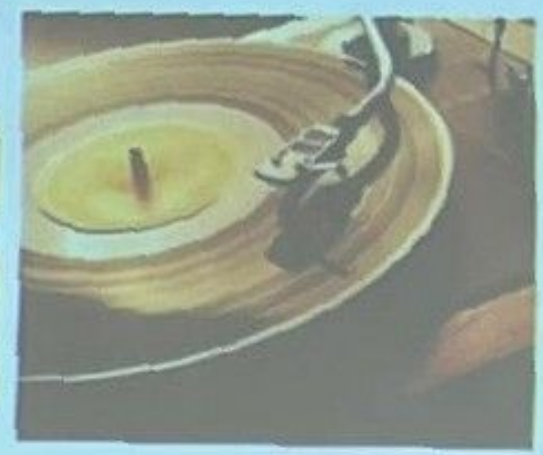


Types of External Memory

- **Magnetic Disk**
  - Removable/Non-Removable
  - RAID
- **Magnetic Tape**
  - Audio Tape/ Cassettes
  - Video Tape/ Cassettes
  - Floppy Disks
  - Photo Reel
- **Optical**
  - CD-ROM
  - CD-Recordable (CD-R)
  - CD-R/W
  - DVD-ROM
  - DVD-R
  - DVD-R/W
- **Solid State**
  - SSD
  - USB Memory Stick

Magnetic Disk

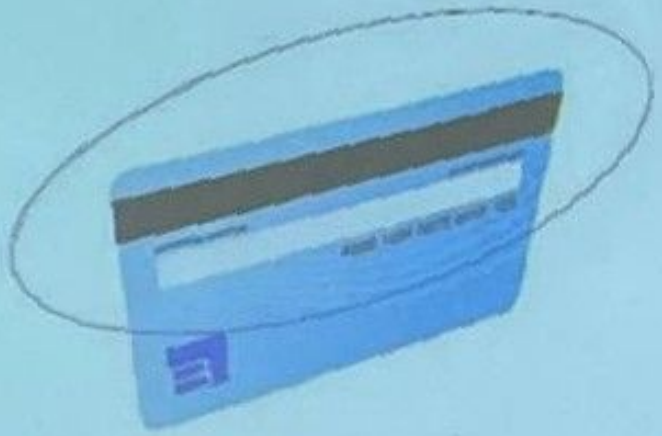
- A magnetic disk is a storage device that can be assumed as the shape of a Gramophone record.
- A magnetic disk is a secondary storage device that uses a magnetization process to write, rewrite and access data.
- This disk is coated on both sides with a thin film of Magnetic material.
- This magnetic material has the property that it can store either '1' or '0' permanently.
- It is covered with a magnetic coating and stores data in the form of tracks, spots and sectors.

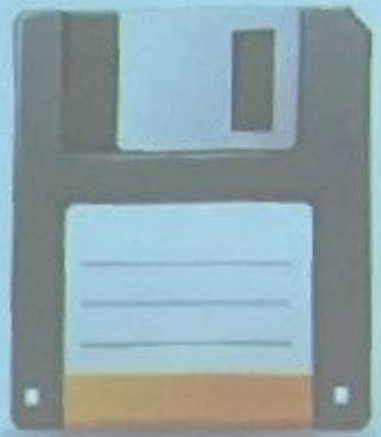


Magnetic Disk Examples

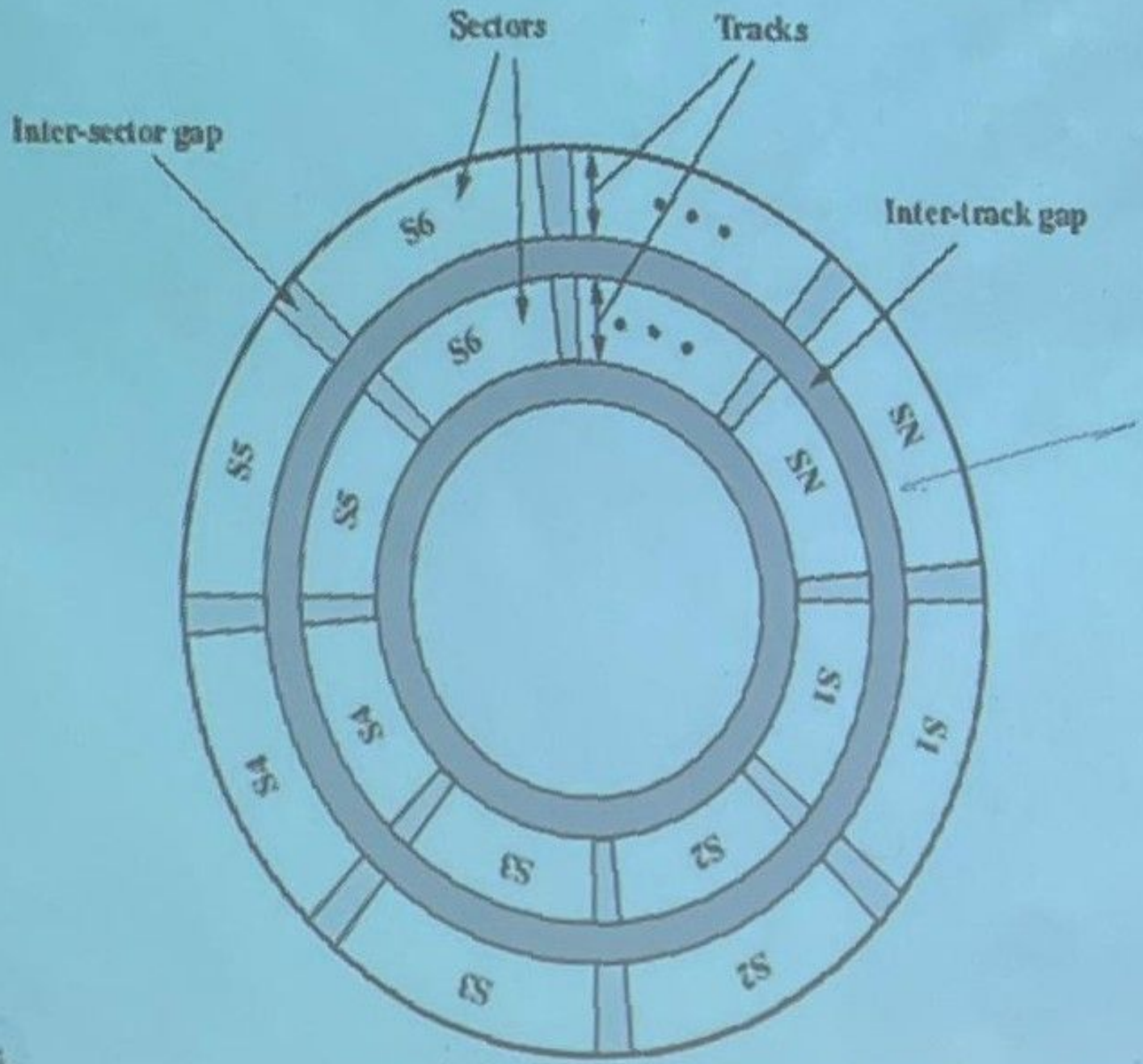
- Primary magnetic storage is Hard Disks used in computers.
- Other magnetic storage media include
  - floppy disks,
  - magnetic tape, and
  - magnetic stripes on credit cards.

Magnetic Disk Examples

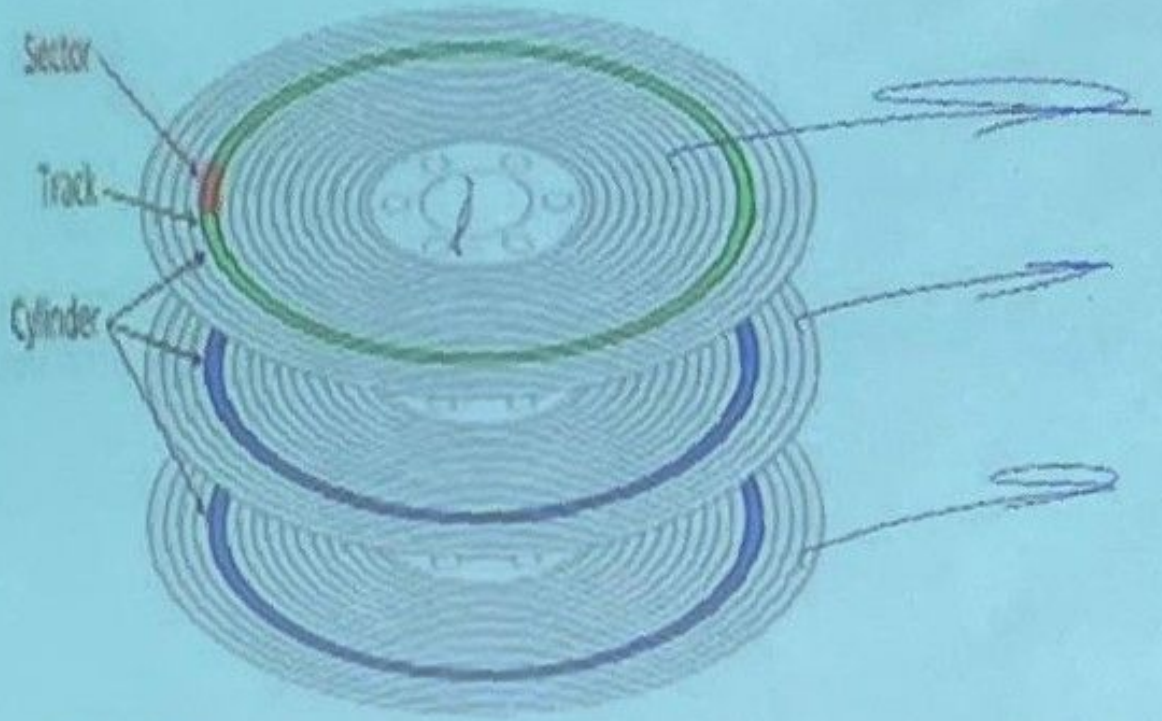


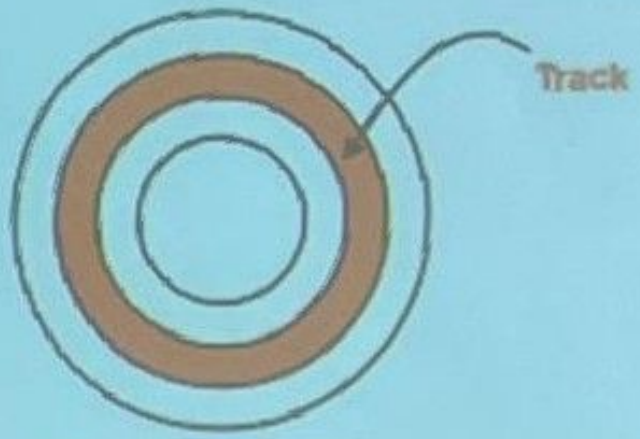


Magnetic Disk Data Layout

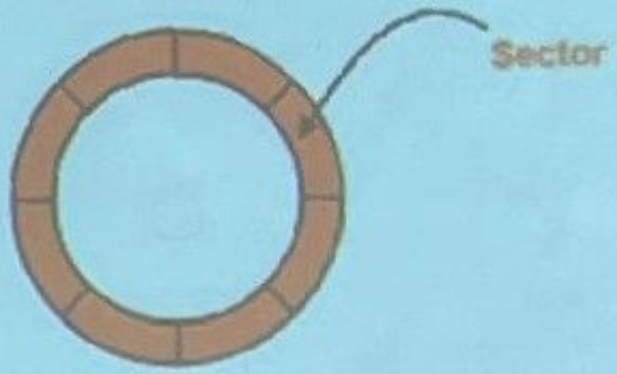


Magnetic Disk Data Layout



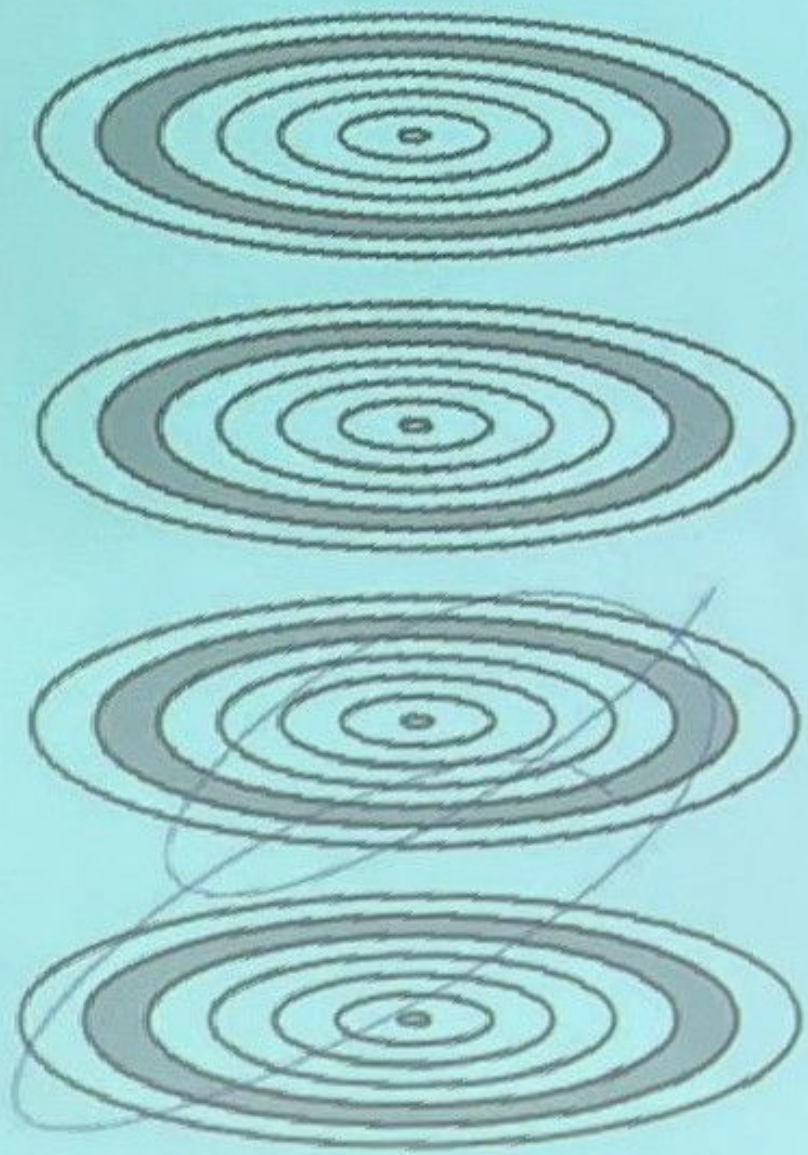


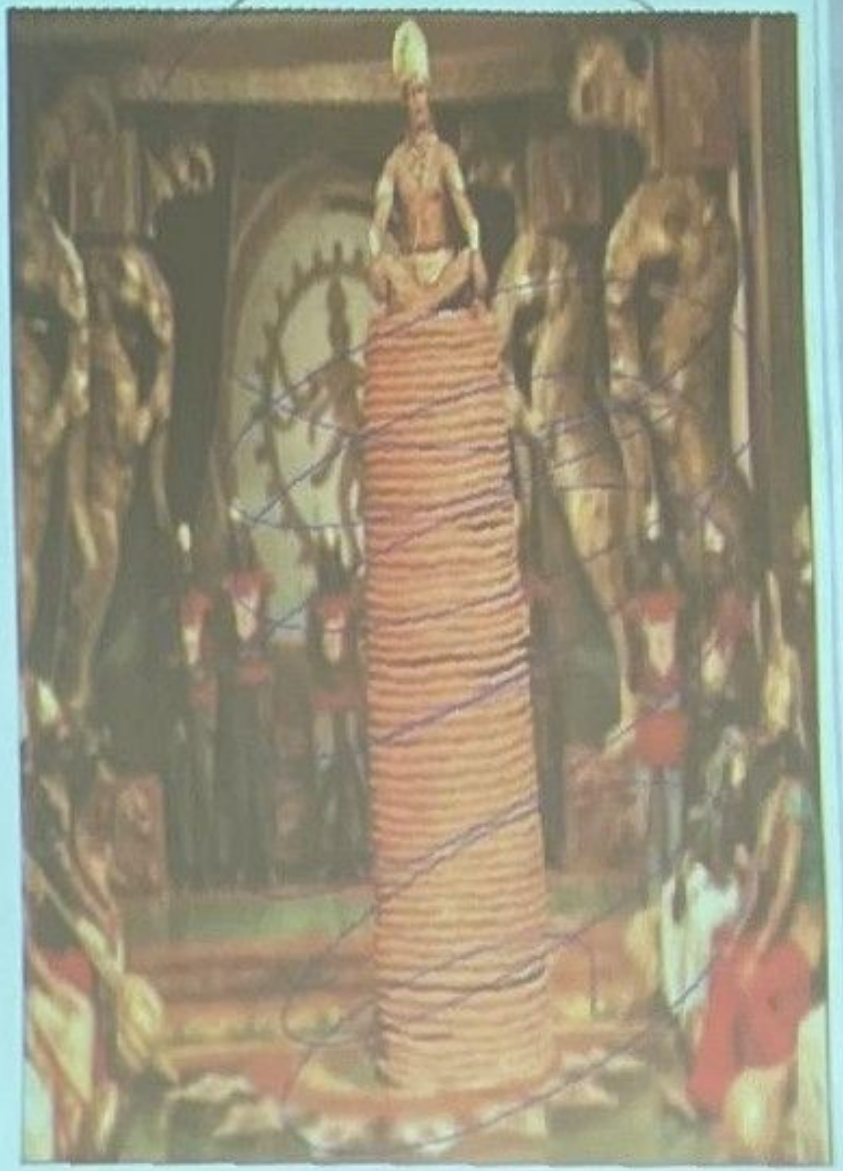
Disk divided into tracks



Track divided into sectors

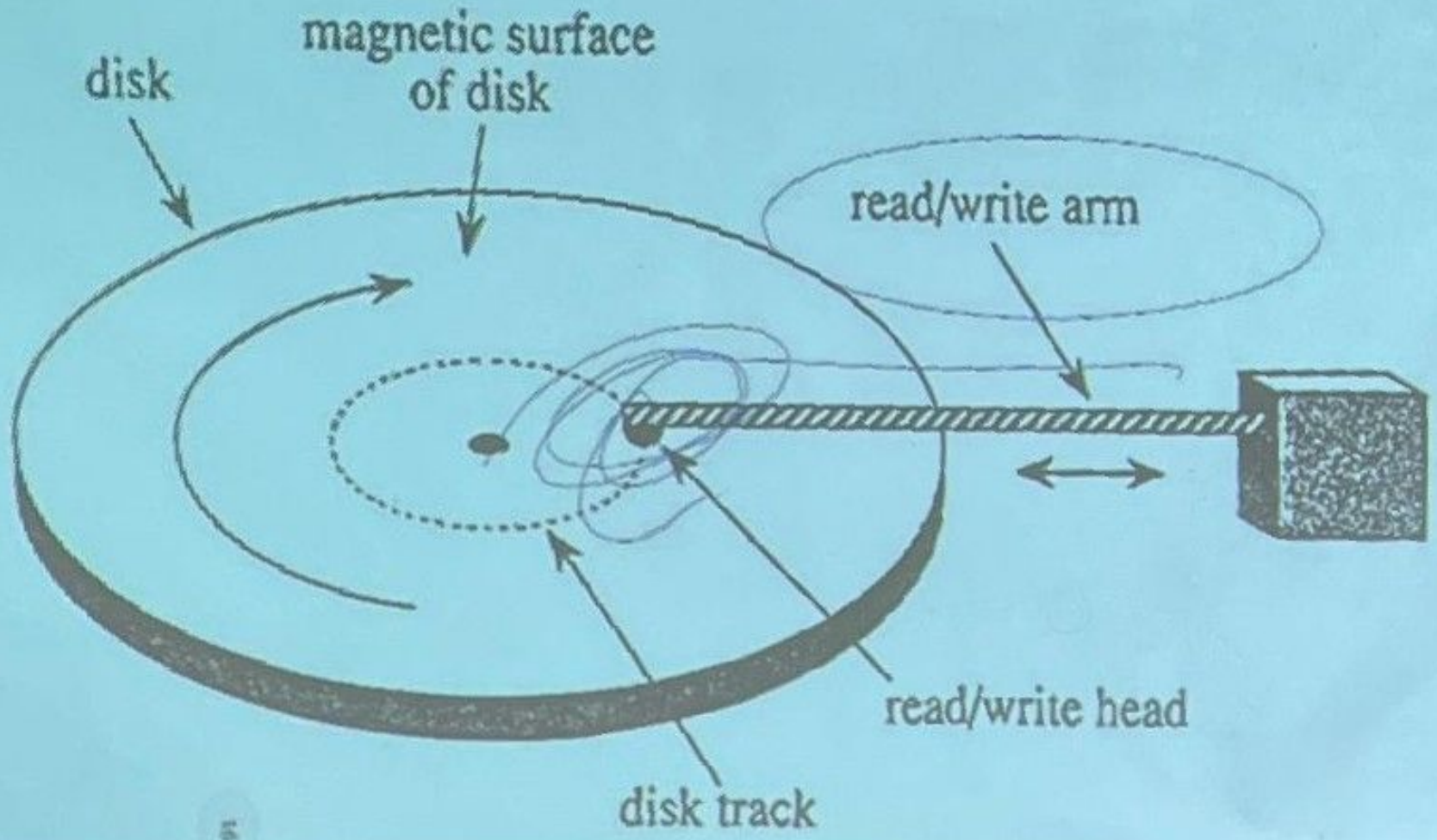
Tracks and Cylinders



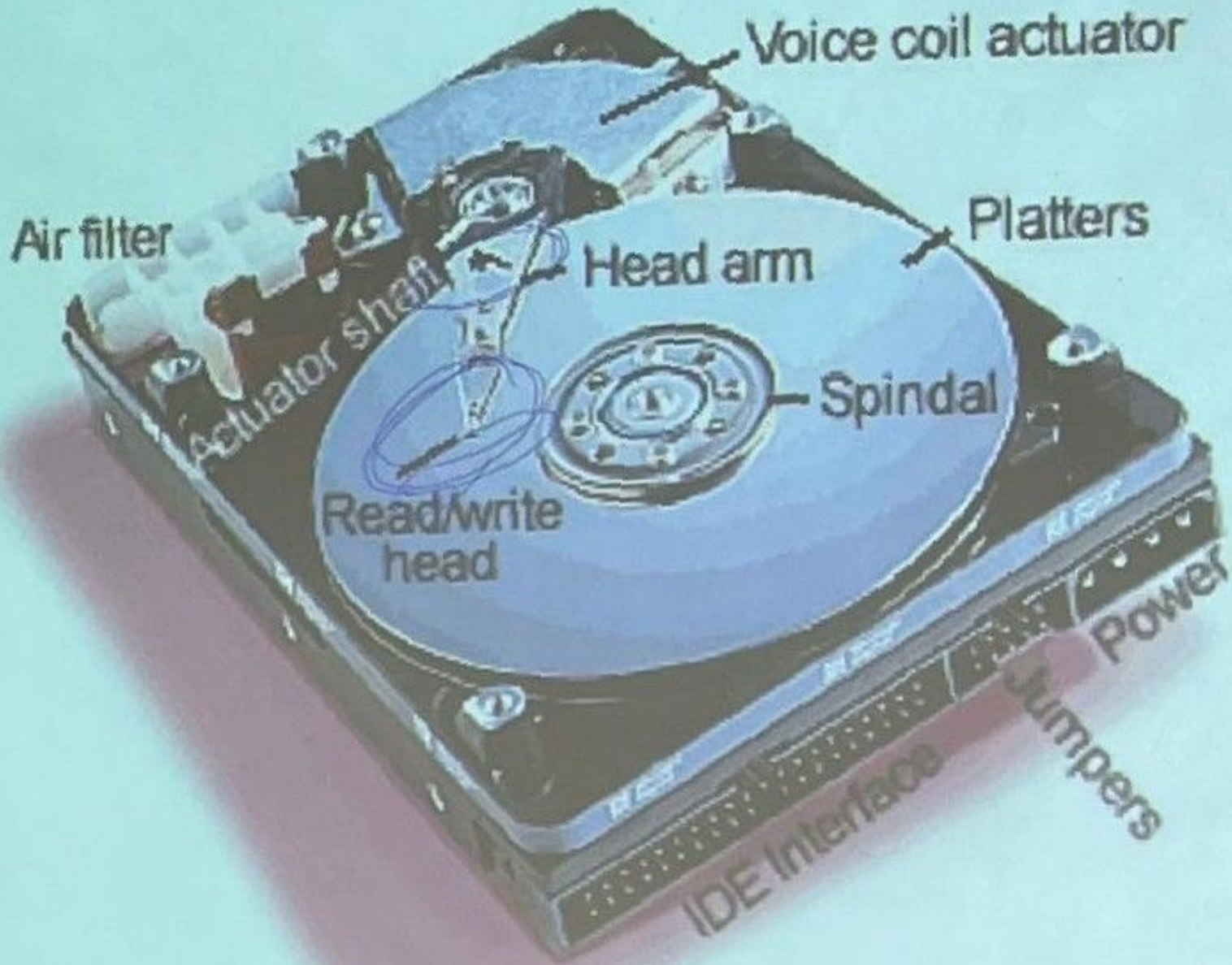


Read and Write Mechanisms

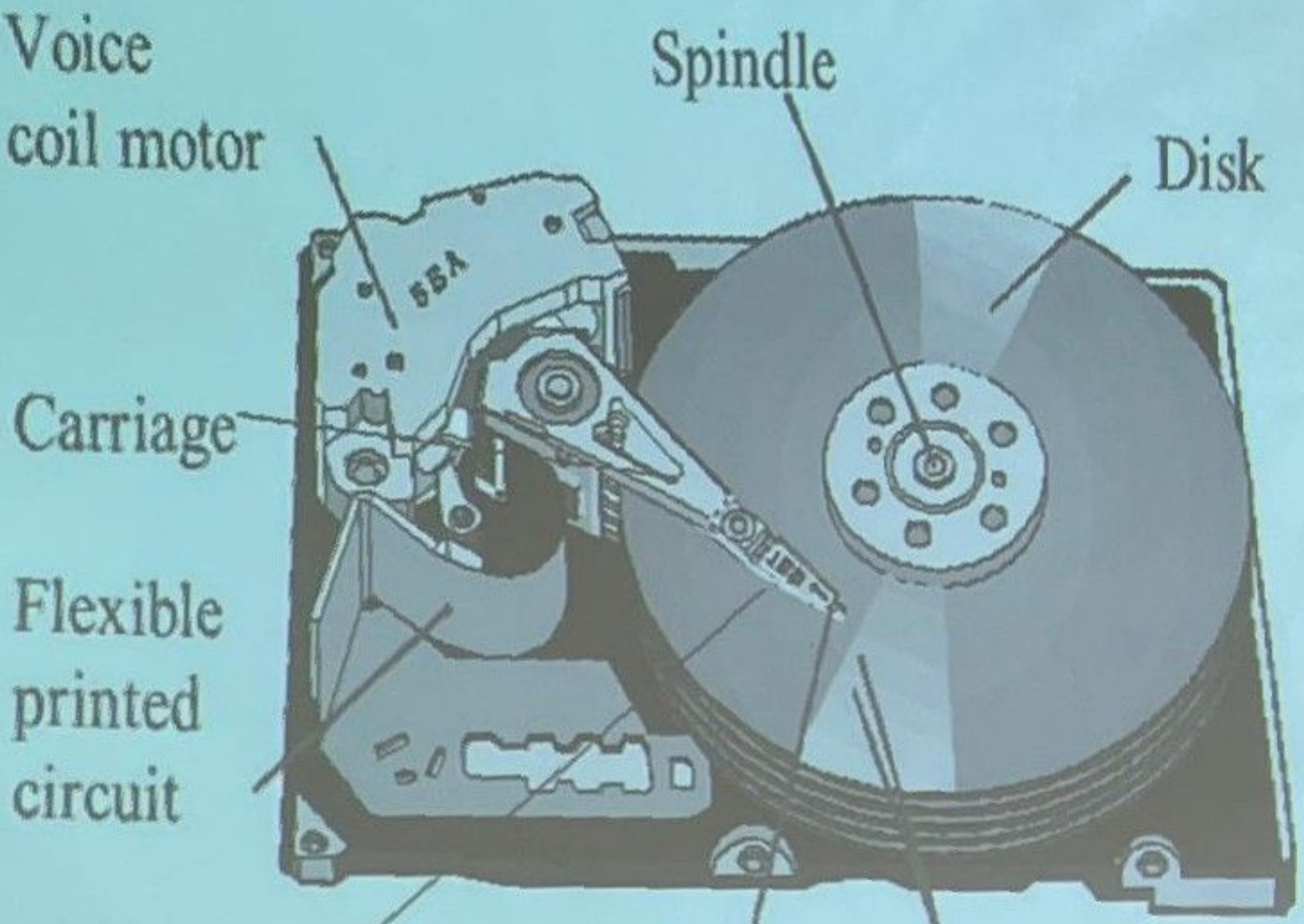
- Recording (write) and retrieval (read) via conductive coil called a head.
- May be single read/write head or separate ones.
- During read/write, head is stationary, platter rotates



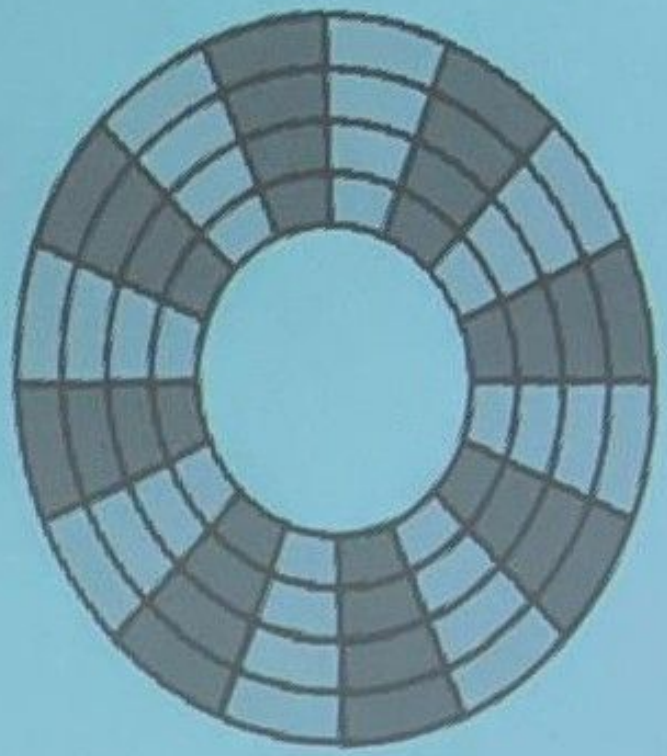
Hard Disk Innards



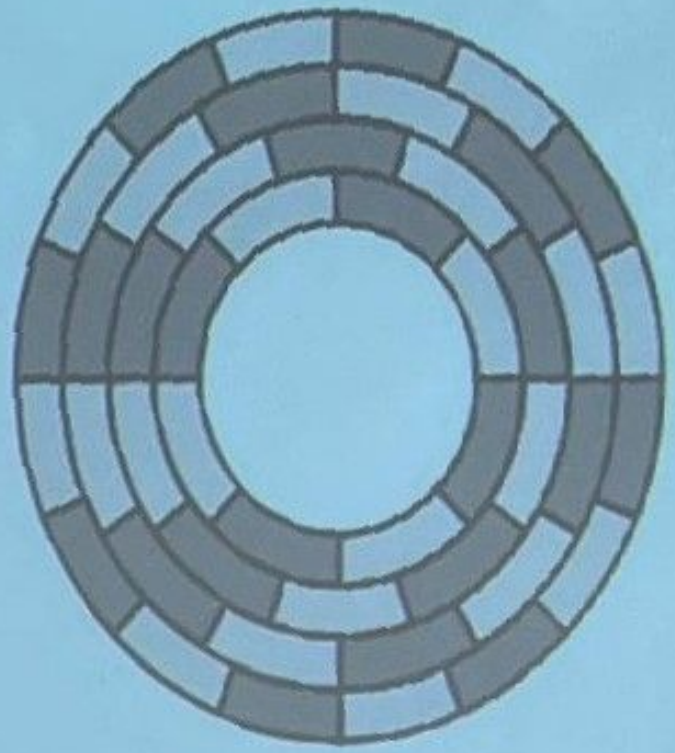
Read and Write Mechanisms



Disk Layout Methods Diagram



(a) Constant angular velocity

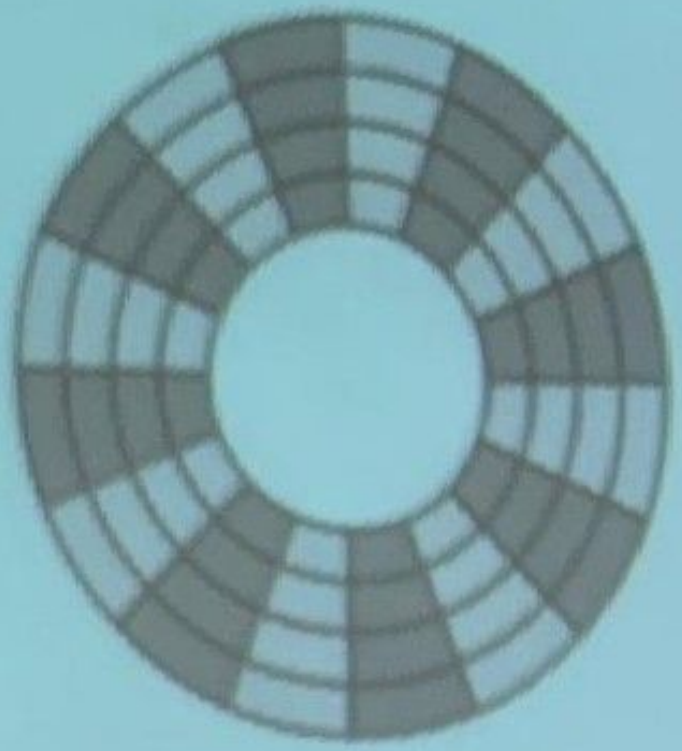


(b) Multiple zoned recording

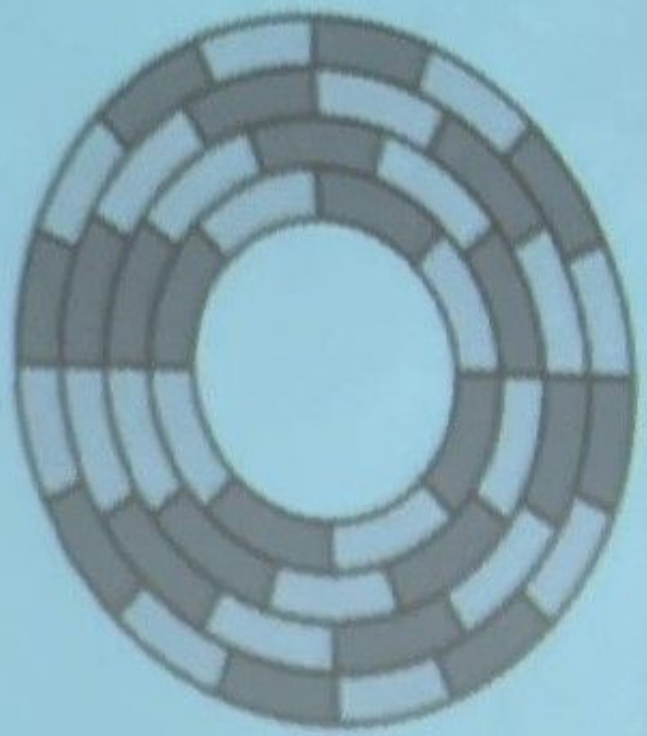
Disk Velocity

- Bit near centre of rotating disk passes fixed point slower than bit on outside of disk
- Increase spacing between bits in different tracks
- Rotate disk at constant angular velocity (CAV)
  - Gives pie shaped sectors and concentric tracks
  - Individual tracks and sectors addressable
  - Move head to given track and wait for given sector
  - Waste of space on outer tracks
    - Lower data density
- Can use zones to increase capacity
  - Each zone has fixed bits per track
  - More complex circuitry

Disk Layout Methods Diagram



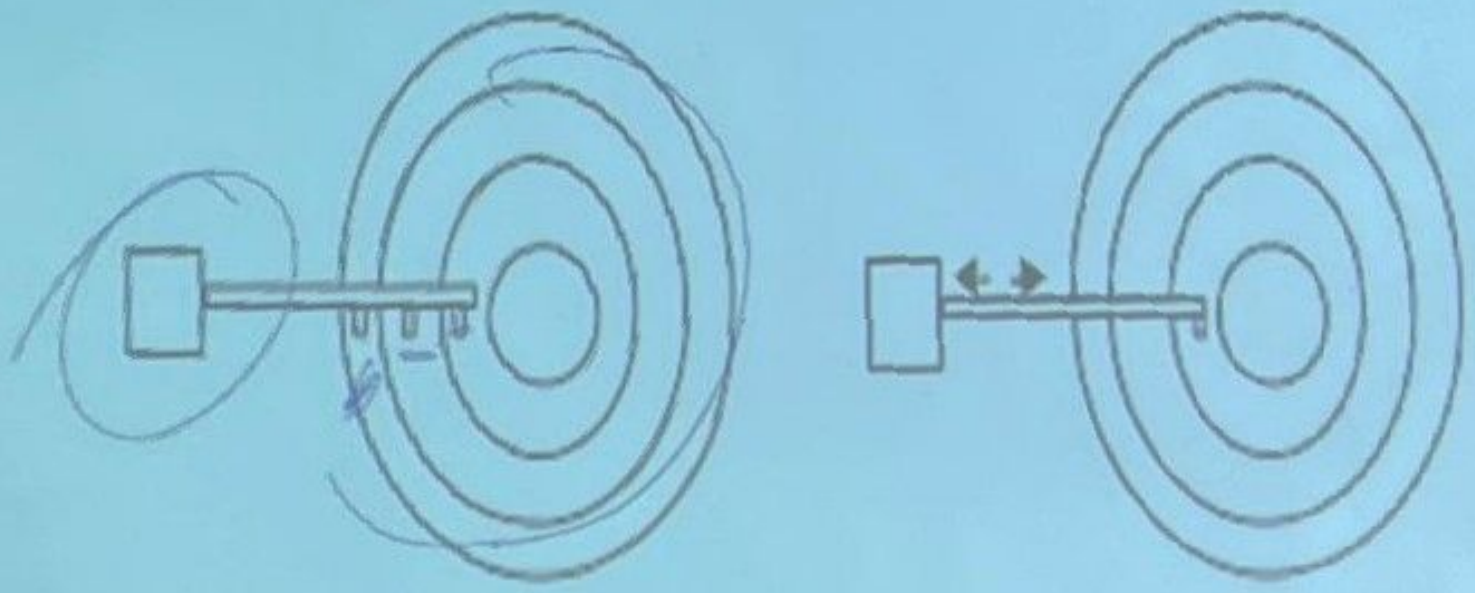
(a) Constant angular velocity



(b) Multiple zoned recording

Fixed Head Disk

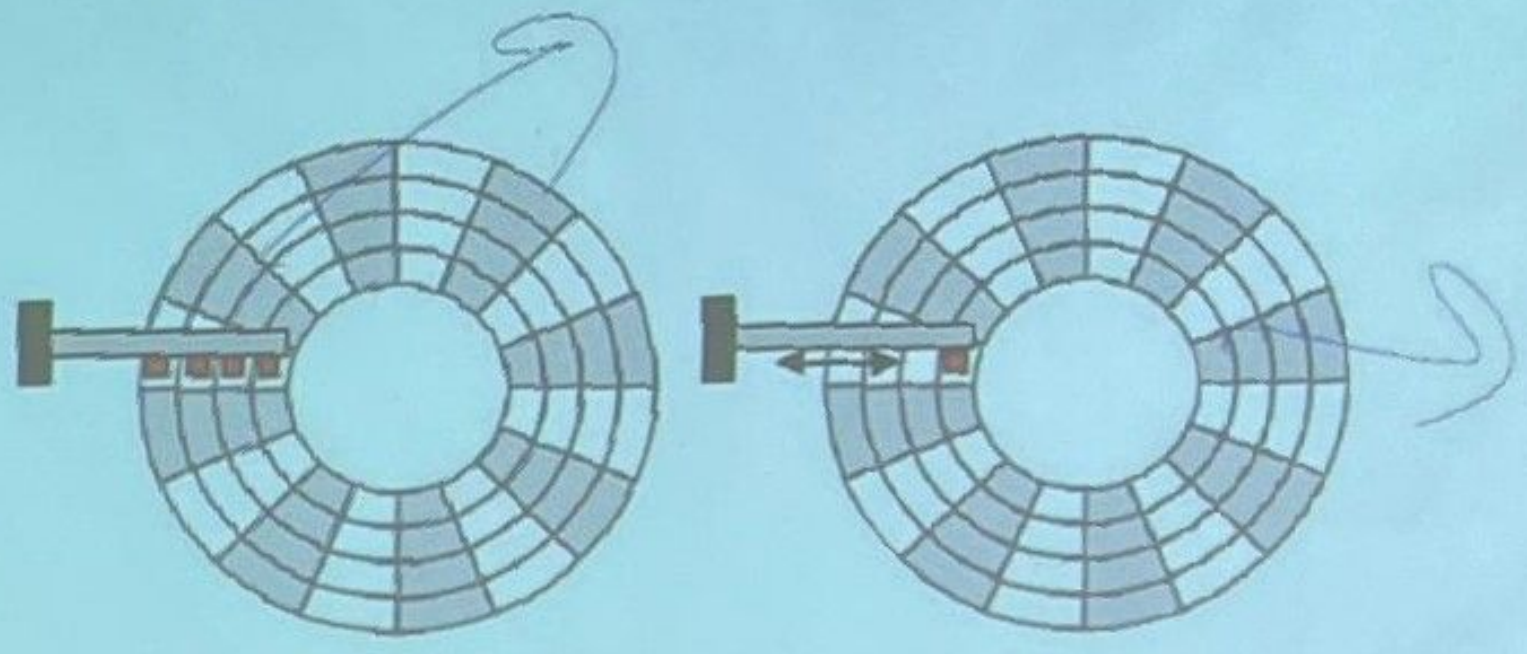
- Fixed head
  - One read write head per track
  - Heads mounted on fixed ridged arm



(a) Fixed head (b) Moveable head

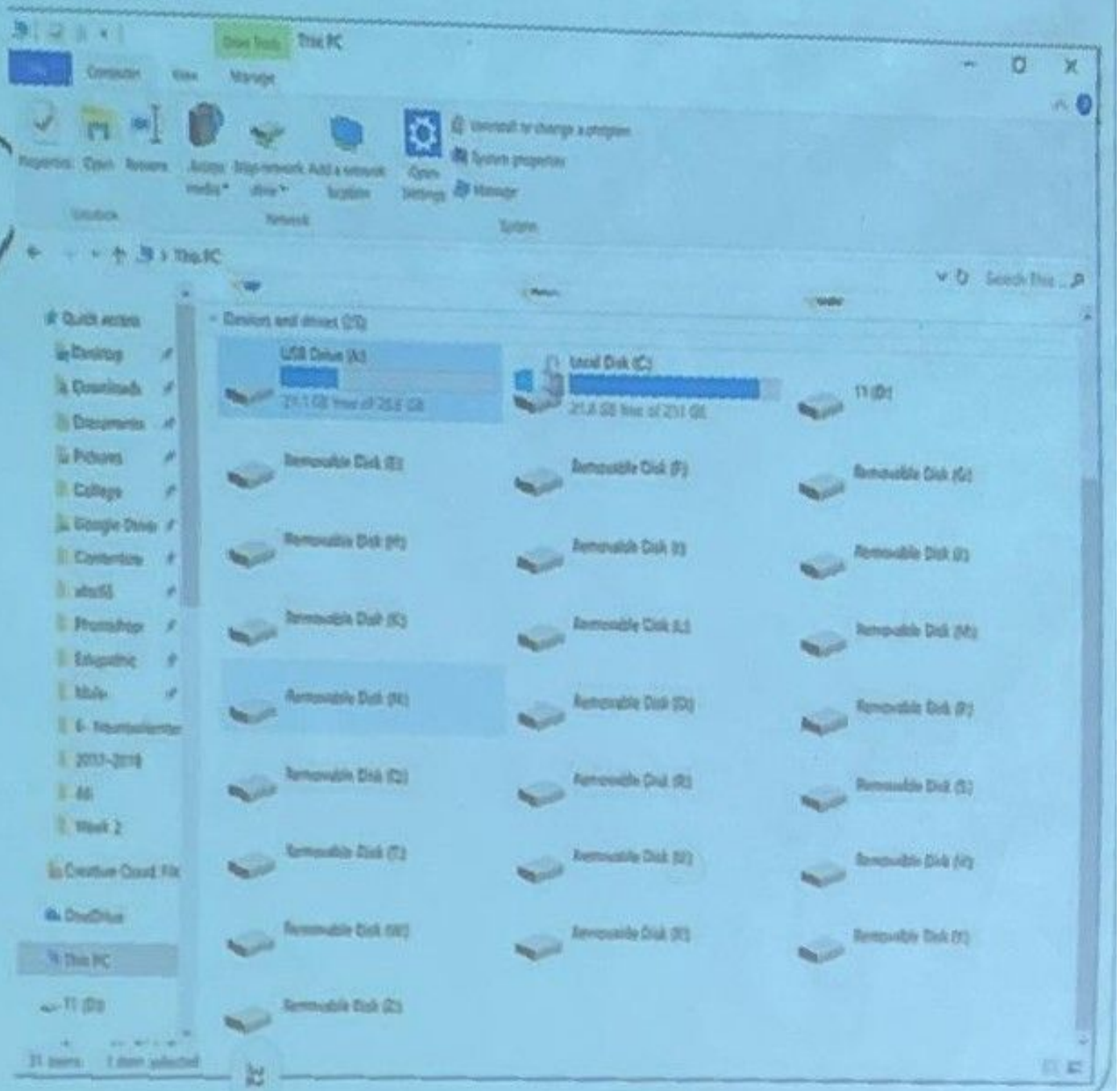
Movable Head Disk

- Movable head
  - One read write head per side
  - Mounted on a movable arm

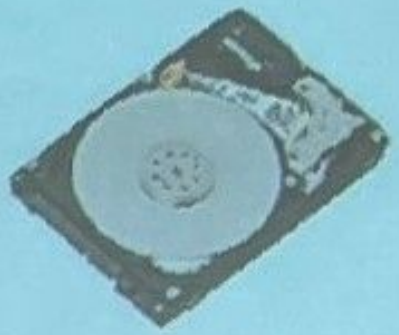


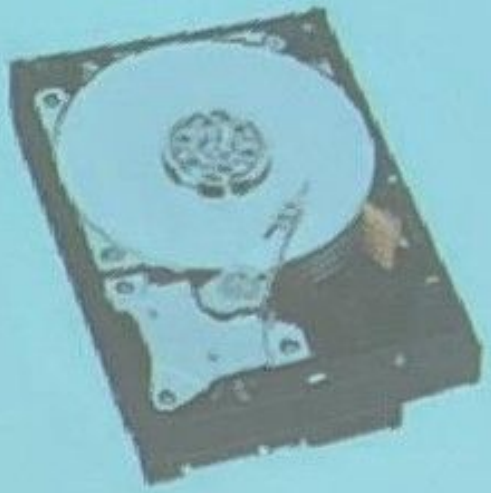
(a) Fixed head (b) Movable head

Removable Disks

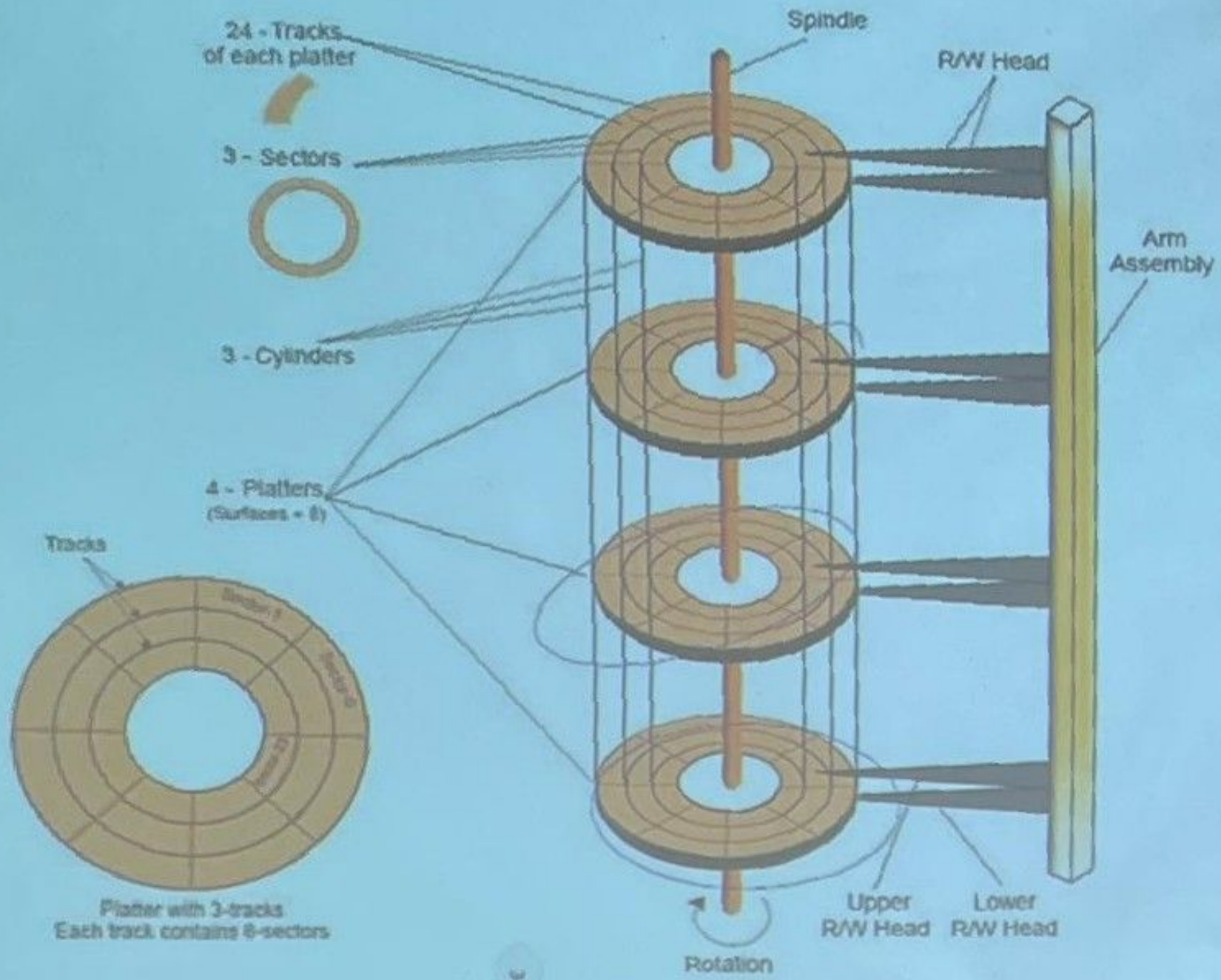


Non-Removable Disks

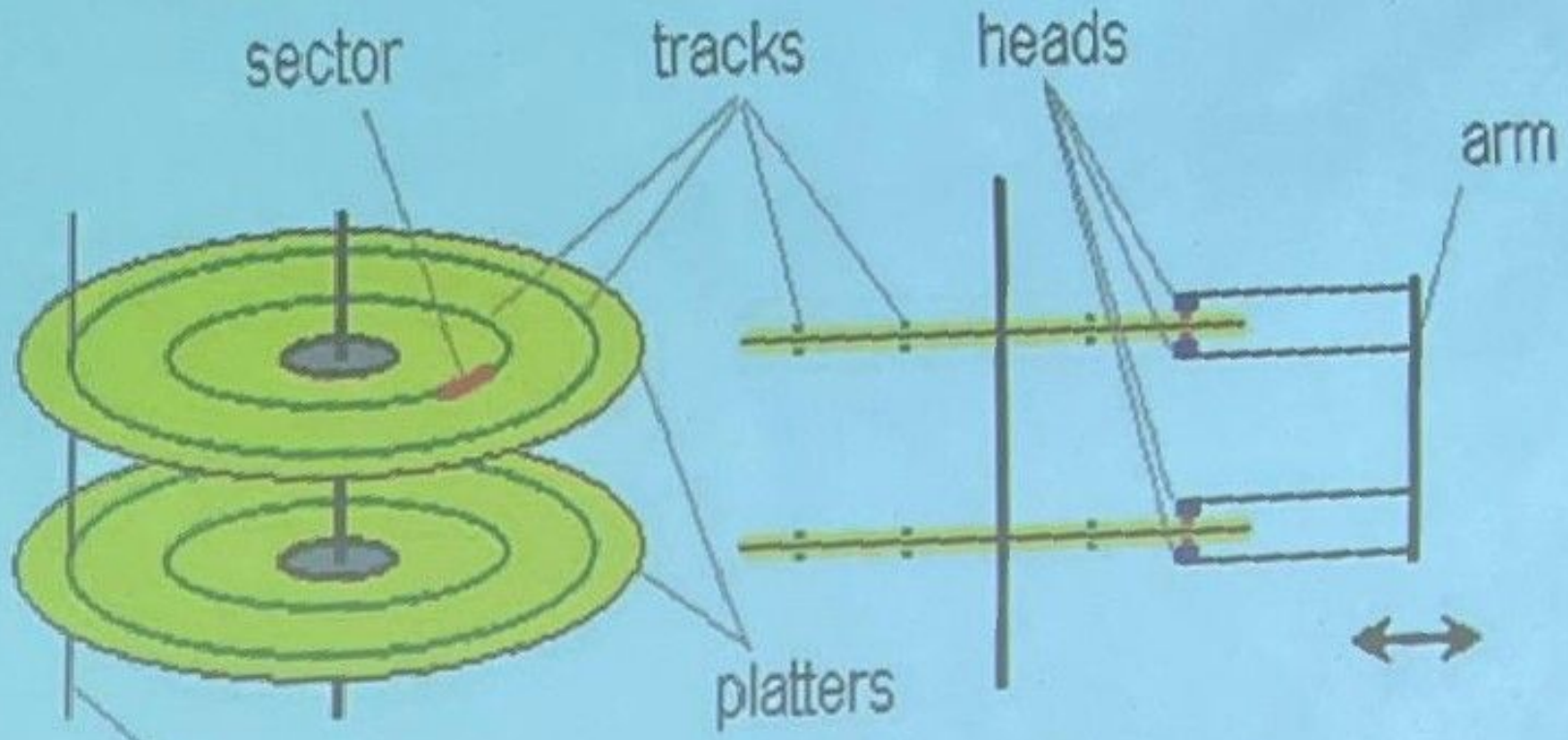




Single/Double Sided R/W Mechanisms



Multiple Platter



cylinder - set of tracks which can be accessed without moving the arm

Speed Parameters

- **Seek time**
  - Moving head to correct track
  - Typical average seek time is under 10 ms
- **(Rotational) latency**
  - Waiting for data to rotate under head
  - 15,000 rpm (rotation per minute) = 250 r/sec
  - 1/250 = 0.004 sec → There is one rotation per 4ms.
  - Thus, average rotation delay (1/2r) : about 2ms
- **Transfer rate**
  - Time required for data transfer
- **Additional delays**
  - Waiting for I/O channels
  - Rotation Positional Sensing (RPS)

seek time
=moving head to correct track
=typical average seek time is under 10ms

(Rotational) latency
=waiting for data to rotate under head
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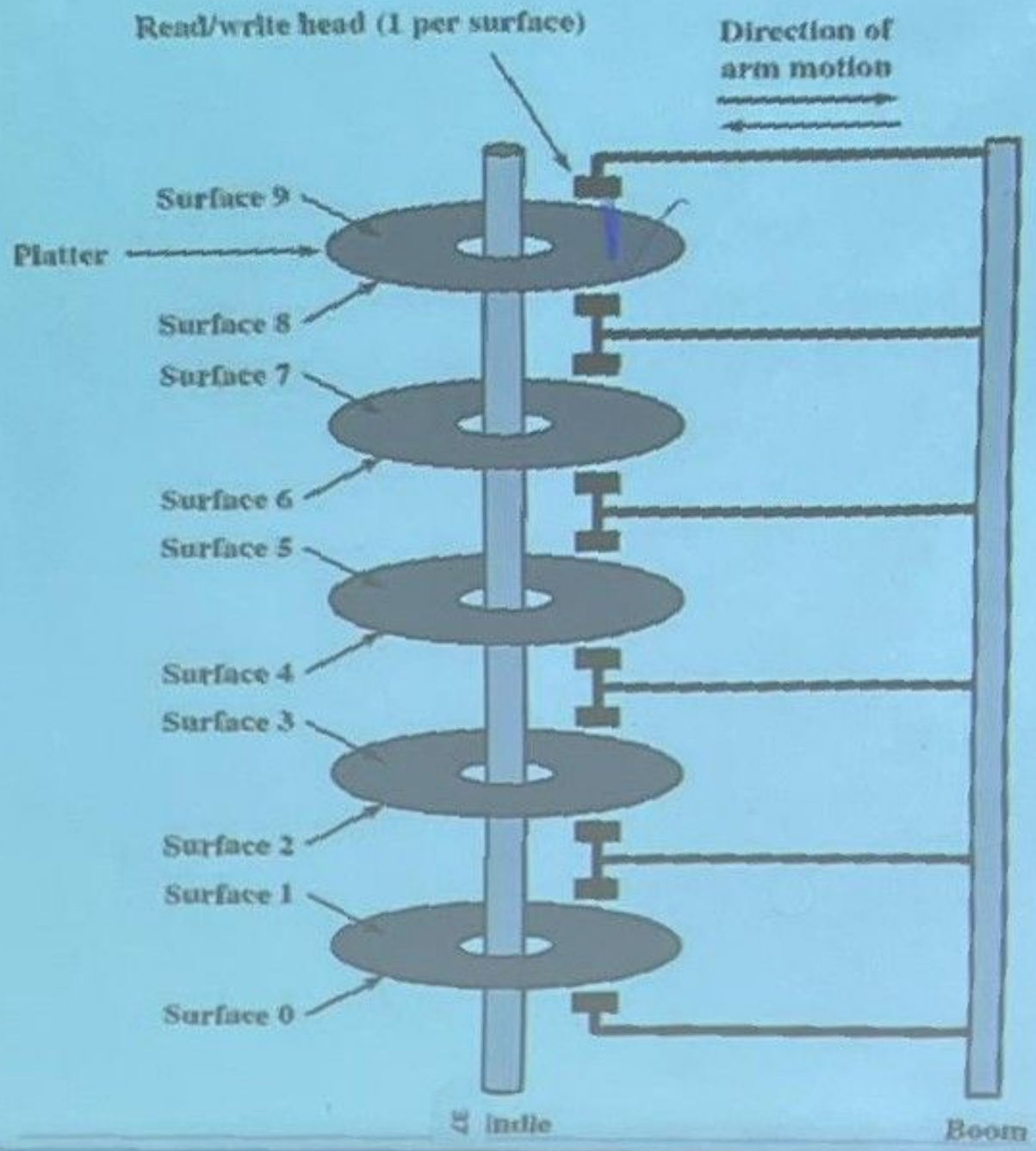
transfer rate
=time required for data transfer

additional delays
=waiting for i/o channels
=Rotor Position Sensor/ Rotation Positional Sensing(RPS)

Multiple Platters



Typical Hard Disk Drive Parameters

Characteristics	Seagate Barracuda 180	Seagate Cheetah X15-36LP	Seagate Barracuda 36ES	Toshiba MK1002MAV	Hitachi Microdrive
Application	High-capacity server	High-performance server	Entry-level desktop	Portable	Handheld devices
Capacity	181.6 GB	36.7 GB	18.4 GB	5 GB	4 GB
Minimum track-to-track seek time	0.8 ms	0.3 ms	1.0 ms	—	1.0 ms
Average seek time	7.4 ms	3.6 ms	9.5 ms	15 ms	12 ms
Spindle speed	7200 rpm	15K rpm	7200	4200 rpm	3600 rpm
Average rotational delay	4.17 ms	2 ms	4.17 ms	7.14 ms	8.33 ms
Maximum transfer rate	160 MB/s	522 to 709 MB/s	25 MB/s	66 MB/s	7.2 MB/s
Bytes per sector	512	512	512	512	512
Sectors per track	793	485	600	63	—
Tracks per cylinder (number of platter surfaces)	24	8	2	2	2
Cylinders (number of tracks on one side of platter)	24,247	18,479	29,851	10,350	—

Speed Parameters

- The **transfer time** to or from the disk depends on the rotation speed of the disk in the following fashion:

$$T = (b/N)*(1/r)$$

  - b : # of bytes to be transferred
  - N : # of bytes on a track
  - r : rotation speed, in revolutions per second

Speed Parameters

- Total average access time

$$Ta = Ts + 1/(2r) + b/(rN)$$

- Ts : average seek time
- 1/(2r) : average rotation delay
- b/(rN) : transfer time